Samsung PRISM GenAI Hackathon 2026 · Theme 02 · Demo round

FixFlow

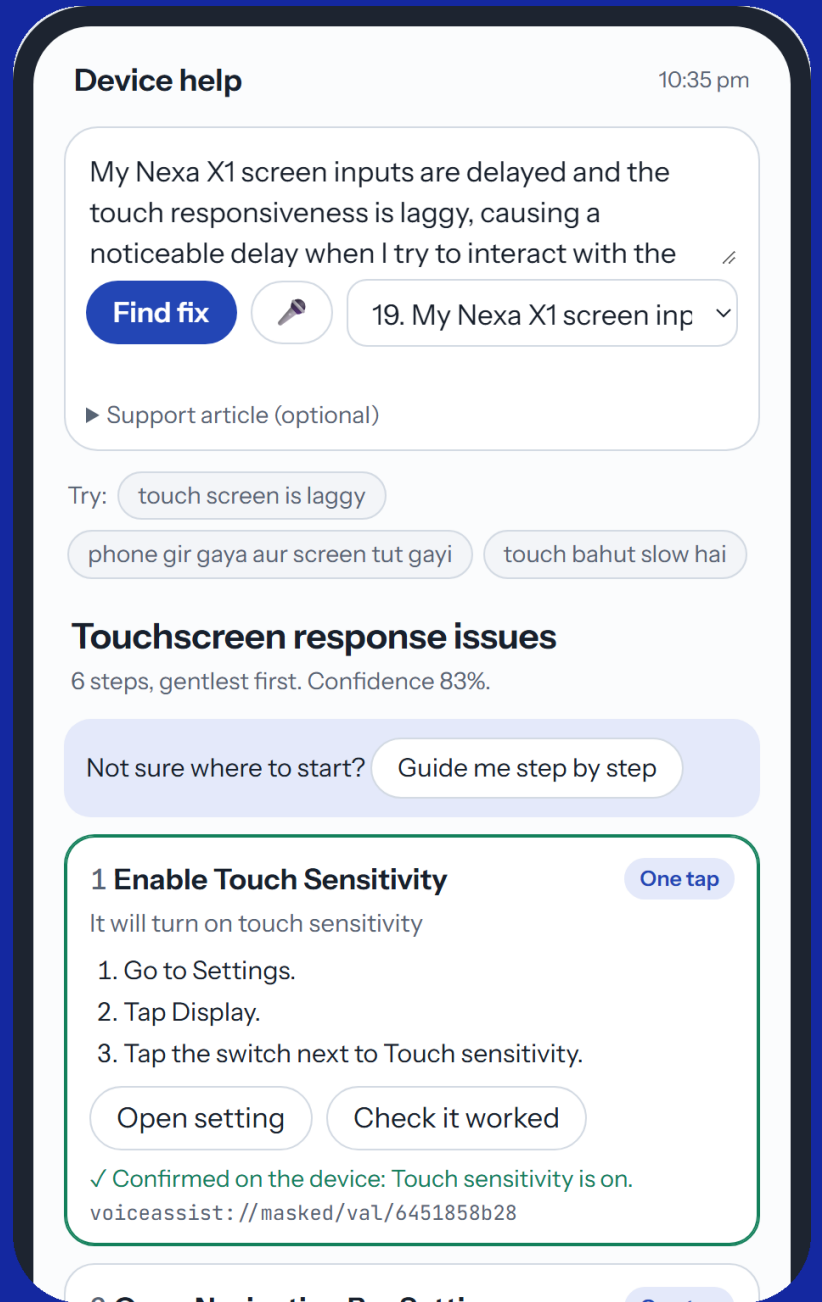
Smart Guided Troubleshooting Engine

A phone complaint in the user's own words goes in. A short, ordered fix plan comes out, with one-tap links into Settings, built only from the support article.

Team SRMIST_QueueQueueing

SRM Institute of Science and Technology

Abhinav Kumar · Abhishek Verma · Ayushi Paul · Ketki Gonnade



Understanding the problem

Turn a vague complaint into a fix the user can tap

WHAT COMES IN

"My Nexa X1 screen inputs are delayed and the touch responsiveness is laggy, causing a noticeable delay when I try to interact with the device."

Official kit, complaint 19

- Its support article (SIIS text)
- A catalog of 578 Settings deeplinks

FixFlow

WHAT MUST GO OUT

Touchscreen response issues

1	Enable Touch Sensitivity	One tap
2	Open Navigation Bar Settings	One tap
3	Check Screen Protectors	By hand
4	Check Usage and Moisture	By hand
5	Check Charger	By hand
6	Restart Your Device	Last resort

Pure JSON, official schema | **Catalog deeplinks only** | **Restart and reset always last** | **No fix in the article: no_match** | **Repeat < 300 ms, new < 8 s**

Caveats identified

Six catches we found in the official kit

1 The article may not fit

Complaint 6 is about a tablet's dark screen; its article is about multi-window. A plan here would be invented.

Extract: relevance check

2 AI models invent steps

An LLM adds tips, settings and web links that the article never mentions.

Extract: citation check

3 Deeplinks are masked codes

Only the descriptions can be matched, and 138 switches come as on/off twins. The wrong twin undoes the fix.

Deeplinks: action-aware search

4 One line, several problems

Complaint 17 holds three complaints in one line. Each needs its own plan.

Understand: problem split

5 Many wordings, one plan

Rewordings and Hinglish must get the same plan, and a repeat must answer in under 300 ms.

Language + Memory

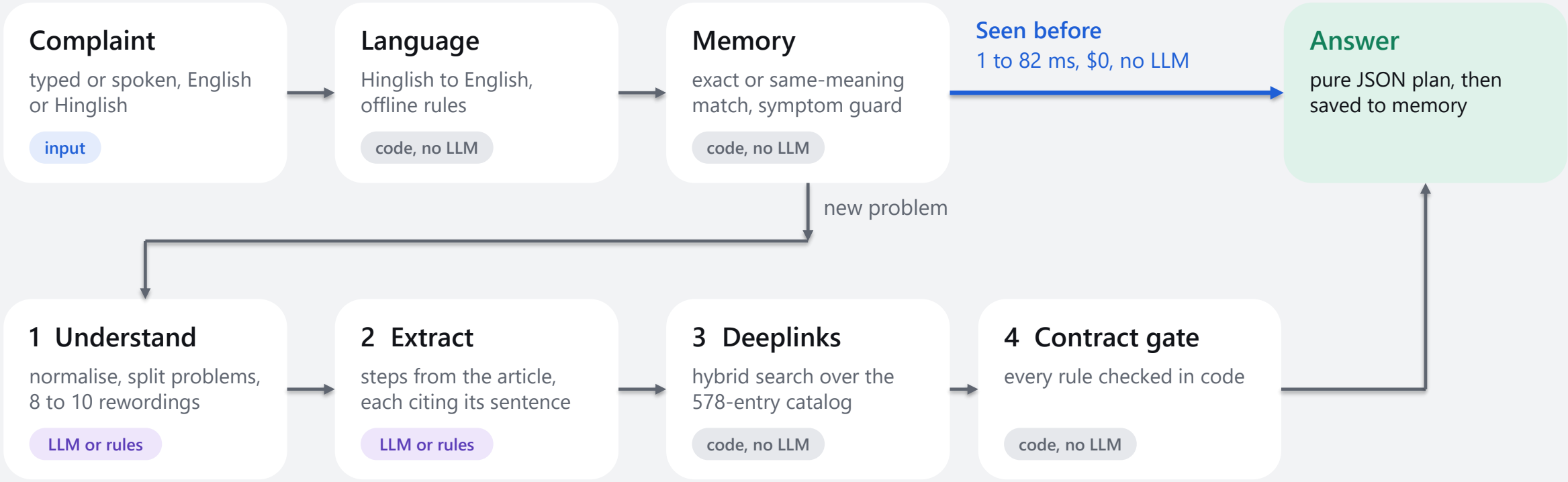
6 The sample breaks the spec

The kit's sample output has 8- and 12-word descriptions; the spec says 5 to 7, starting “It will”.

Contract gate: we follow the spec

Core architecture

Memory first. A new problem passes four checked stages



Runs with no AI key

Every LLM stage has a rules path. CPU only, one Docker image.

Gemini is optional

Flash-Lite with a model fallback chain and a 7 s budget per call, then rules.

Same meaning, same plan

Temperature 0, fixed seed, and memory return identical plans.

Core technology modules

Three core modules keep every answer grounded

Grounded extraction

The article is split into numbered sentences. Every step must cite them, and code deletes any step whose words are not in the sentences it cites.

2.60 / 3

step accuracy on the official lines; 2.73 on held-out scenarios

Deeplink mapper

Keyword (BM25) plus meaning search over catalog descriptions, never the masked code. It picks on vs off and the exact screen, then copies the entry verbatim.

100%

link precision. LLM picking links: 75%, keyword rules: 58%

Guarded memory

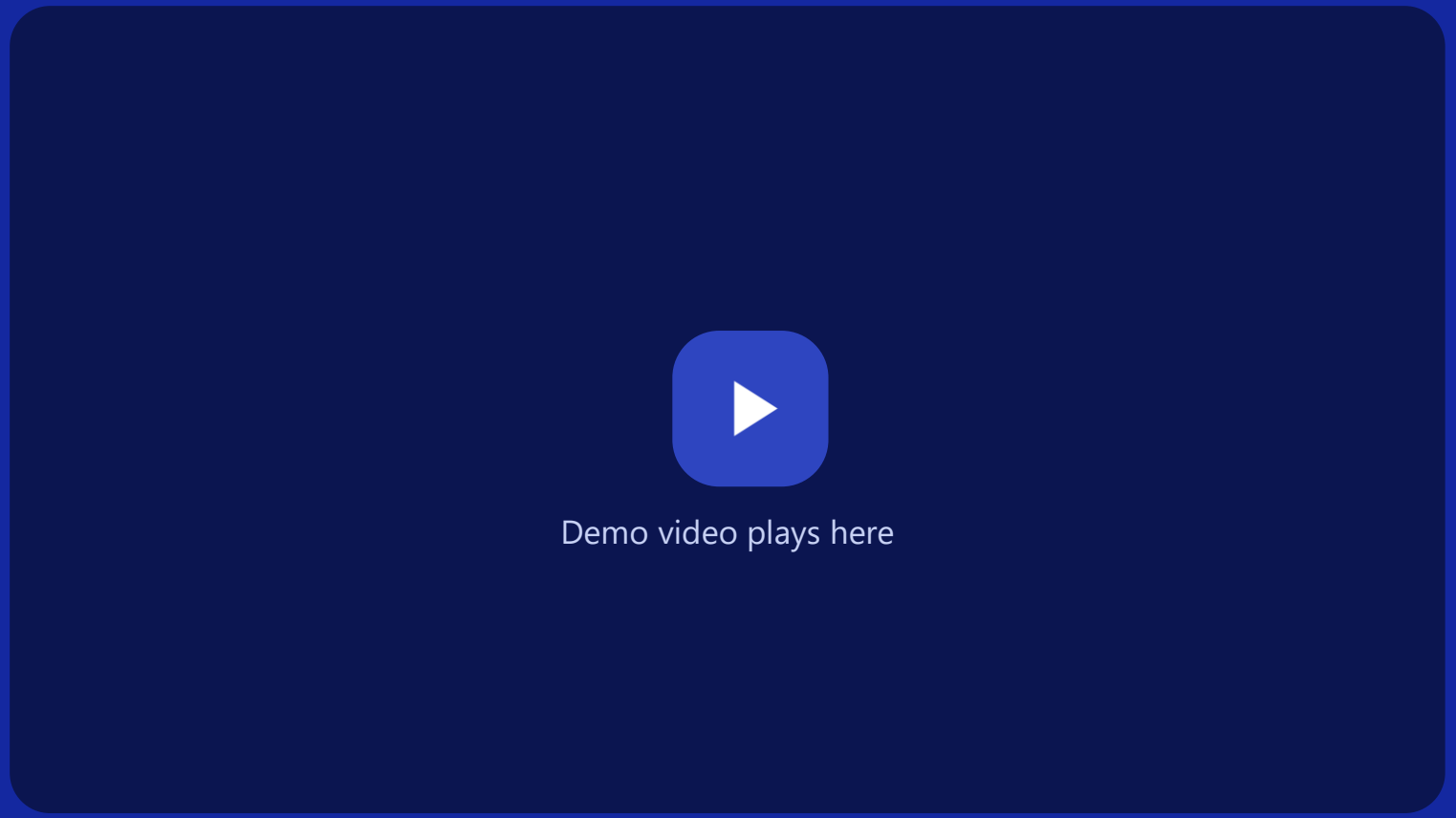
Each plan is stored with its complaint and 8 to 10 rewordings. A hit needs close meaning and the same symptom, so a flicker complaint never gets the cracked-screen plan.

87.5%

of held-out rewordings get the same plan; 0 wrong hits

Contract gate in code: 100% schema-valid, 100% rule compliance and 0 web links on the official, dev and held-out sets

FixFlow in under two minutes

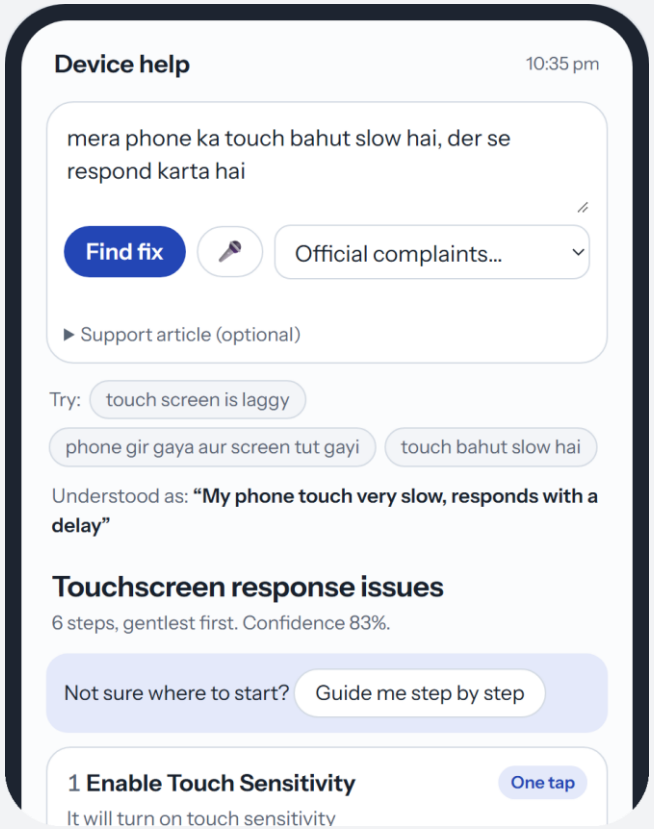


IN THE VIDEO

0:00	An official complaint becomes a 6-step plan. One tap opens the setting, and its check confirms it.
0:30	The same problem in new words: same plan, from memory, in about 20 ms.
0:50	A Hinglish complaint, shown as “Understood as”, gets the same plan.
1:10	An article that does not fit: no_match, nothing invented.
1:25	A new problem is taught with one article; a rewording is then answered from memory.

Extension for Samsung

Built for the way Samsung users in India ask for help



Hinglish in, same plan out, 27 ms

Hinglish, understood offline

"mera phone ka touch bahut slow hai" is read as English and gets the touchscreen plan. Our Hinglish set: right plan 3 of 15 before, 11 of 15 now, 0 wrong. Devanagari Hindi goes through Gemini.

Tap the fix, then verify it

One-tap steps carry the catalog's deeplink plus its validation deeplink (570 of 578 entries have one), so the device can open the screen and confirm the change. The demo UI simulates the device.

Voice and step-by-step guidance

The complaint can be spoken (browser speech), and "Guide me step by step" walks through one step at a time until it is fixed.

No complaint text has to leave the server

Rules mode needs no API key and no GPU: one Docker image on a CPU. Outputs stay brand-neutral, checked by a test.

Scalability

A new problem needs one article and zero code

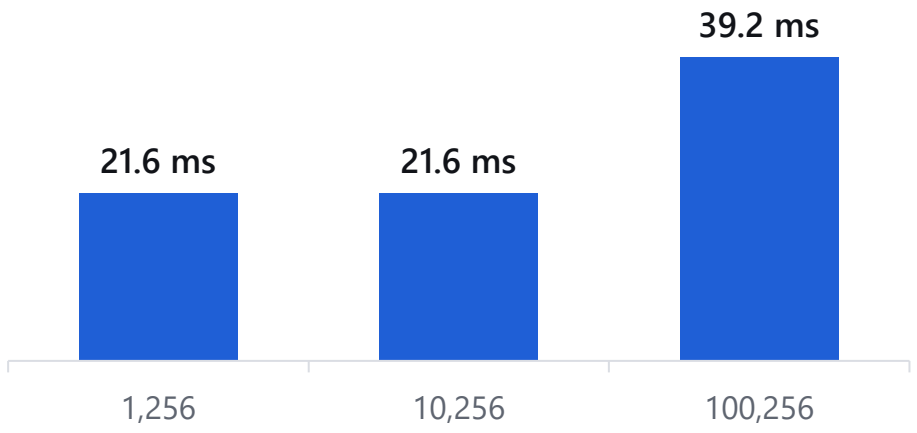
Add the support article
No code, no retraining

Compile it once
Rules 0.3 s, $0. Gemini 2.6 s, $0.0004

Answer every rewording
From memory, P95 under 50 ms, $0

Memory lookup stays fast as it grows

P95 lookup time (ms) by number of stored problems



8 / 8

new problem types got a plan with 0 code changes (battery, charging, Bluetooth, camera)

7 / 8

rewordings of the new problems got the right plan; 0 false hits on 5 unrelated

555 / 556

phone Settings screens in the catalog reached by search, 0 wrong

98 req/s

with 50 users at once, 0 errors, P95 642 ms, one CPU worker

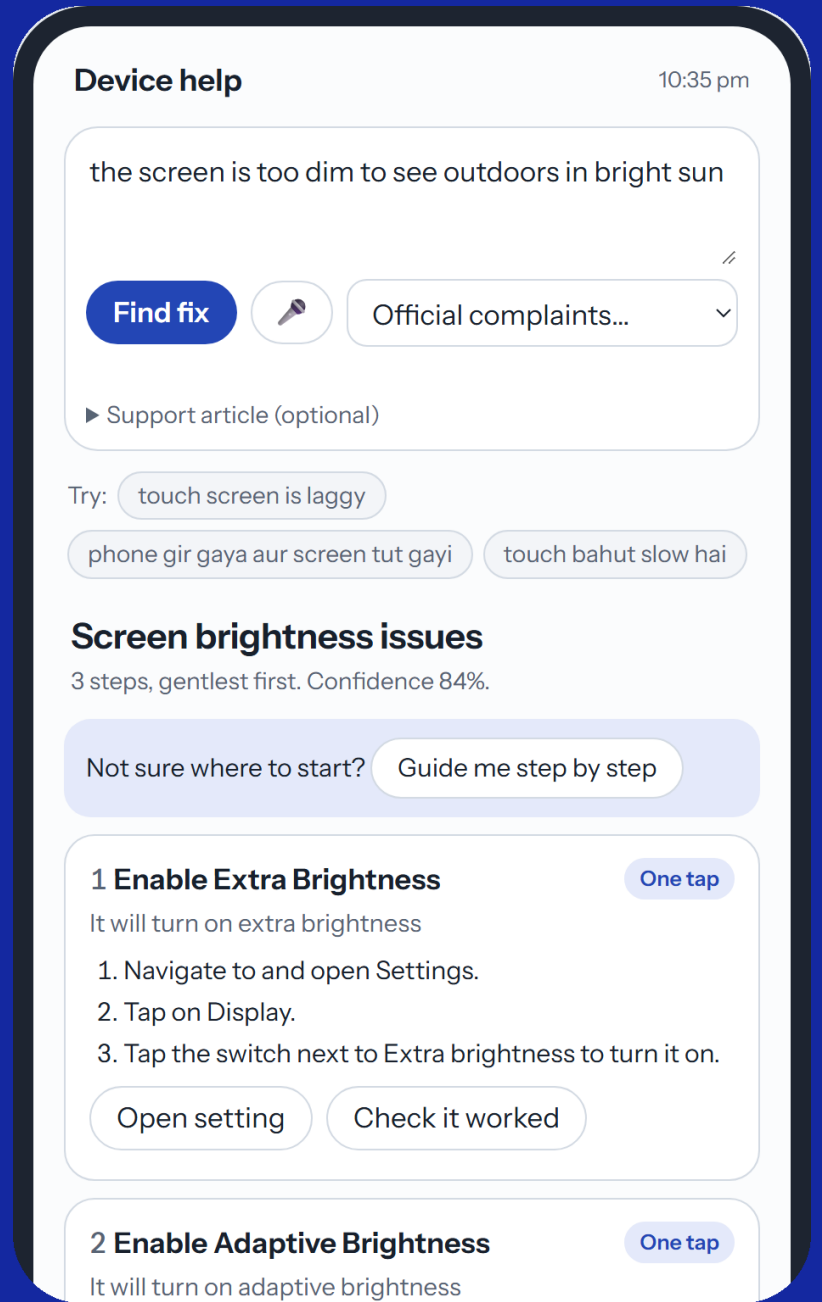
Thank you

We are happy to take your questions.

- **Grounded** every step cites the support article
- **Exact** every deeplink is copied from the catalog
- **Fast** repeats come from memory in milliseconds, at $0
- **Scalable** a new problem is one article, no code

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Backup for Q&A

Results on the official and held-out sets

Metric (target)	Official D1	Held-out D2b
Schema-valid / rule compliance (99% / 95%)	100% / 100%	100% / 100%
Web links in output (0)	0	0
Settings steps with a catalog deeplink (90%)	100%	100%
Step accuracy, gold proxy (0 to 3)	2.60	2.73
Step accuracy, LLM judge (0 to 3)	1.94	2.06
Deeplink relevance (0 to 2) / precision	2.00 / 100%	1.82 / 91%
Says no_match at the right time	95%	90%

Deeplink mapping ablation

Same extraction, only the mapper changes (D1 + D2, on gold)

Hybrid BM25 + dense (ours)

100% precision relevance 1.88 cost $0

Full-LLM mapping

75% precision relevance 1.76 cost $0.00125

Keyword rules

58% precision relevance 1.29 cost $0

Backup for Q&A

Known limits, and what the engine does about them

Limit

Hardest unseen rewordings: 75% hits

On the final never-tuned set D3c, 75% hits and 62.5% right plan (target 80%). Two look-alike black-screen complaints got the flicker plan. Reported as is; the held-out set D3b reaches 91.7% / 87.5%.

Limit

Gemini free tier is often overloaded

503 and 429 errors are common. The adapter tries other models, then the rules path answers inside the budget. Every request still returns a valid plan (14 of 32 cold runs used the LLM).

Limit

Answer key is AI-assisted

Gold labels for the official lines were written with AI help; a human check is pending. An independent LLM judge is reported next to them.

Limit

Some article mismatches slip through

One official line pairs a TV aspect-ratio article with a phone complaint; the LLM still builds a plan there.